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Technical Presentation Report

Topic : Comparative Analysis of IaaS, PaaS, and SaaS

Course: Cloud Computing and Big Data Analytics (CSA1504)

Objective

To compare Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) in terms of control, flexibility, cost, and practical use cases for educational institutions.

Introduction

Cloud computing delivers computing resources such as servers, storage, databases, networking, and software over the internet. It enables organizations to access scalable IT resources without maintaining their own infrastructure.

Definitions

IaaS: Provides virtualized computing resources including virtual machines, storage, and networking.

PaaS: Provides a complete platform for developing, testing, and deploying applications.

SaaS: Provides ready-to-use software applications accessed through a web browser.

Architecture

Users

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SaaS (Applications)

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PaaS (Runtime, Database, Development Tools)

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IaaS (Virtual Machines, Storage, Networking)

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Physical Data Center managed by Cloud Provider

Comparison

Feature	IaaS	PaaS	SaaS
Control	High	Medium	Low

Flexibility	High	Medium	Low
Cost	Pay-as-you-go	Moderate	Subscription
Best for	IT Admins	Developers	End Users

Real-World Use Case

An educational institution can use:

- IaaS for virtual computer laboratories and backup servers.
- PaaS for student programming projects and application deployment.
- SaaS for email, online learning, and office productivity tools such as Google Workspace and Microsoft 365.

Industry Relevance

Cloud service models are widely adopted because they reduce infrastructure costs, improve scalability, simplify maintenance, and support remote access.

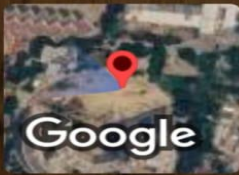
Conclusion

IaaS offers maximum control, PaaS simplifies application development, and SaaS provides ready-to-use software. Educational institutions benefit by combining all three models according to their requirements.



Module 1: Patient Application & Smart Appointment System

- User Registration & Login - Secure account creation and authentication for patients.
- Profile Management - Manage personal information and medical details.
- Hospital & Doctor Search - Search hospitals, departments, and available doctors.
- Appointment Booking - Schedule appointments with AI-recommended time slots.
- Queue & Notifications - Track live queue status and receive real-time appointment updates.
- Medical Services - Access digital prescriptions, medical records, QR check-in, and online payment.



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